

Exercise Class 3 — Uncertainty (MWG Ch. 6) and the Past-Exam Hour

Advanced Microeconomics 1 · Thursday 29 October, 11:00–13:00

Format (120 min). *Hour 1:* Problems 1–5 at the board (assignments posted Monday 26 October); Problem 5 is the hard item and the instructor closes it. *Hour 2:* past-exam questions spanning Ch. 1–3, 5 and 6, worked under time pressure — real past papers are on the course page; Problems 6–7 below are prelim-format stand-ins in case you have exhausted them. Bring the synopsis.

Hour 1 — Chapter 6

Problem 1 The expected utility theorem, small

Let $C = \{c_1, c_2, c_3\}$ with $c_1 \succ c_2 \succ c_3$, and let $\succsim$ on lotteries be rational, continuous, and satisfy independence.

- (a) Show that for $\beta, \gamma \in [0, 1]$: $\beta\delta_{c_1} + (1 - \beta)\delta_{c_3} \succ \gamma\delta_{c_1} + (1 - \gamma)\delta_{c_3}$ iff $\beta > \gamma$, using independence only. (δ_c = the sure outcome c .)
- (b) Define u_2 by $\delta_{c_2} \sim u_2\delta_{c_1} + (1 - u_2)\delta_{c_3}$ (why does a unique such u_2 exist?). Setting $u_1 = 1$, $u_3 = 0$, prove that $L \succsim L' \iff \sum_n p_n u_n \geq \sum_n p'_n u_n$ for all lotteries on C .
- (c) Show directly that $\tilde{u}_n = \beta u_n + \gamma$ with $\beta > 0$ represents the same preferences, and that no other transformation of (u_1, u_2, u_3) does.

Problem 2 Allais, formally

With prizes (\$5m, \$1m, \$0), consider $L_1 = (0, 1, 0)$, $L'_1 = (.10, .89, .01)$, $L_2 = (0, .11, .89)$, $L'_2 = (.10, 0, .90)$.

- (a) Show that the modal pattern $L_1 \succ L'_1$ and $L'_2 \succ L_2$ is inconsistent with *any* expected-utility preference: write both rankings as inequalities in (u_{5m}, u_{1m}, u_0) and derive a contradiction.
- (b) Exhibit the common-consequence structure: find lotteries L, L', L'' and α such that the two pairs are $\alpha L + (1 - \alpha)L''$ -type mixtures differing only in L'' .
- (c) In the Marschak–Machina triangle (probability of \$0 on the horizontal axis, of \$5m on the vertical), draw the four lotteries and show the modal pattern requires indifference lines that fan out.

Problem 3 CARA, CRRA, and the Arrow–Pratt approximation

- (a) For $u(x) = -e^{-ax}$, compute the certainty equivalent of $x \sim N(\mu, \sigma^2)$ exactly. Show the risk premium is $\frac{1}{2}a\sigma^2$, independent of wealth, and reconcile with $\pi \approx \frac{1}{2}\sigma^2 r_A$.
- (b) For $u(x) = x^{1-\sigma}/(1-\sigma)$, compute the certainty equivalent of the 50–50 gamble over $\{w(1+k), w(1-k)\}$ for $\sigma = 1$ (take $u = \ln$), $\sigma = 2$, and $\sigma = 10$, with $k = 0.1$. Express as a share of w . What do the magnitudes tell you about calibrating σ from large risks?
- (c) Derive the approximation $\pi \approx \frac{1}{2}\sigma^2 r_A(w)$ from the definition $u(w - \pi) = \mathbb{E}u(w + \varepsilon)$ by second-order Taylor expansion, stating exactly where mean-zero and “small” are used.

Problem 4 Comparative risk aversion and the portfolio problem

Wealth w is split between a safe asset (return 0) and a risky asset with excess return z , $\mathbb{E}z > 0$: terminal wealth $w + \alpha z$.

- (a) Show that any risk-averse EU maximizer with $\mathbb{E}z > 0$ invests $\alpha^* > 0$: risk aversion never explains *zero* participation in a favorable gamble. (Differentiate at $\alpha = 0$.)
- (b) Show that if agent 2 is more risk-averse than agent 1 in the Pratt sense ($u_2 = \psi(u_1)$, ψ concave), then $\alpha_2^* \leq \alpha_1^*$.
- (c) Under DARA, show $d\alpha^*/dw \geq 0$: the risky asset is a normal good.

Problem 5 Hard item: SOSD, mean-preserving spreads, and the variance trap

- (a) Let G be obtained from F by a mean-preserving spread. Prove $\int u dG \leq \int u dF$ for every concave u (work with the conditional-mean construction: $y = x + \varepsilon$, $\mathbb{E}[\varepsilon | x] = 0$, and Jensen).
- (b) Prove the integral condition: same means and $\int_{-\infty}^x [G(t) - F(t)] dt \geq 0$ for all x implies every concave nondecreasing u prefers F (integrate by parts, twice).
- (c) **Construct** two lotteries with equal means where $\text{Var}(G) > \text{Var}(F)$ but G is *not* SOSD-dominated by F — and exhibit a concave u preferring G . (Hint: put G ’s extra variance in a thin far-right tail, where a piecewise-linear concave u with an early kink doesn’t feel it; spread F symmetrically through the region where the kink bites.)
- (d) Moral, in two sentences: when is mean–variance analysis exactly right, and what does it silently assume otherwise?

Hour 2 — Prelim-format stand-ins

Problem 6 (45 min format) Consumer theory through one functional form

Let $e(p_1, p_2, u) = u p_1^\alpha p_2^{1-\alpha}$, $\alpha \in (0, 1)$.

- (a) Verify e has all the properties of an expenditure function. Recover $h(p, u)$, $v(p, w)$, $x(p, w)$, and the direct utility.

- (b) Compute the Slutsky matrix two ways (from $D_p h$ and from $D_p x + D_w x x^\top$) and verify all three structural properties.
- (c) A tax raises p_1 by 10%. Give CV and EV in closed form; which is larger and why? For which parameter values is ΔCS a good approximation?
- (d) You observe two choices: $x = (2, 1)$ at $p = (1, 2)$ and $x' = (1, 2)$ at $p' = (2, 1)$, same wealth. Are they consistent with *this* model? With *any* utility maximization? (Check GARP.)

Problem 7 (45 min format) A firm and its risks

A price-taking firm has technology $q = \sqrt{z}$, input price $w = 1$.

- (a) Derive $c(q)$, supply $q(p)$, and $\pi(p)$; verify Hotelling's lemma.
- (b) Output price is random: $p \in \{p_L, p_H\}$ with probabilities $(\frac{1}{2}, \frac{1}{2})$, chosen *after* p is observed. Show expected profit is convex in p and conclude the firm *likes* price risk (Oi 1961). Reconcile with risk aversion: who exactly is assumed risk-neutral here, and why did 5.G warn you about this?
- (c) Now the owner is an expected-utility maximizer with strictly concave u consuming the profit, and the firm must commit to z *before* p is realized. Write the FOC. Show the chosen z is below the certainty benchmark at $\bar{p} = \mathbb{E}p$ if u exhibits DARA and profit is increasing in p . Interpret: when does “maximize expected profit” stop being what owners want?